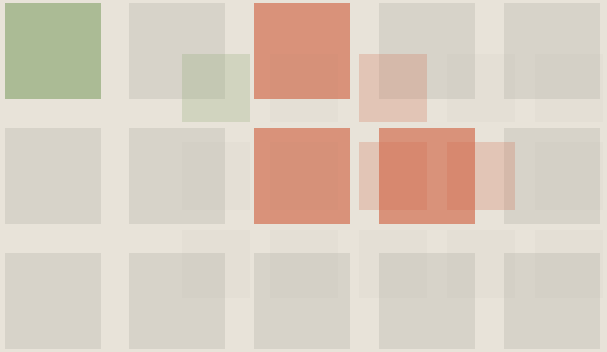
# Anomaly detection for data quality, measured inside the ETL pipeline

Progress review at the end of M1. Rule-based and classical machine-learning detectors run through one pipeline and are compared per anomaly type, on a held-out split, over five seeds.



| | |
|---|---|
| Author | Theo Bigand · 2026-09-01 |
| Repository | github.com/theobigand/etl-anomaly-benchmark |
| Code | ~1.7k LOC · 82 tests · CI green |
| Datasets | Credit Card Fraud · Cardio · Water Quality · synthetic, 6 anomaly types |

02 THE QUESTION

# Does machine learning in an ETL pipeline beat plain rules, and for which anomalies?

The industry default is rule-based: auditable, cheap, and blind to what the author did not anticipate. Recent work adds ML detectors inside the flow, but the evidence is thin, one algorithm on one dataset. This thesis builds the missing head-to-head benchmark and breaks the result down by anomaly type.

What generic anomaly benchmarks leave out

**EVALUATED IN A VACUUM**

Algorithms scored outside any pipeline, no integration cost measured.

**NO DECLARATIVE BASELINE**

The rules that dominate production are never in the comparison.

**ONE AGGREGATE SCORE PER DATASET**

Hiding which family is good at which kind of defect.

**A SINGLE FLAG THRESHOLD**

Reported as one operating point, not the precision-recall trade-off.

03 THE ORIGINAL CONTRIBUTION

# A controlled dataset that labels six anomaly types separately

12,360 records generated with Faker and NumPy, six types injected at 3% each, every row tagged with its category. Recall **and** precision are reported per type. The magnitudes are one **severity** multiplier, so the choice is swept rather than fixed.

**6**

anomaly types, labelled and measured separately

| | |
|---|---|
| MISSING VALUE | one or two feature cells set to null |
| EXACT DUPLICATE | a clean row copied, same business attributes |
| UNIVARIATE OUTLIER | one numeric field pushed far out of range |
| MULTIVARIATE OUTLIER | every field in range, the combination not (an 18-year-old with a top-percentile balance) |
| TEMPORAL DRIFT | recent-signup rows whose income and balance distribution has shifted up |
| INVALID FORMAT | malformed e-mail, out-of-domain country code, impossible age |

Because the category of every injected row is known, the benchmark reports a recall **and** a precision per type: the part that generic anomaly benchmarks, which publish one aggregate score, do not provide.

04 THE SYSTEM

# One interface, one measured path

The demo and every benchmark call the same `detect()` stage, so latency is the real in-pipeline fit + predict + score time. A continuous `score()` on top of the boolean `predict()` makes ROC-AUC and PR-AUC computable; `detect(fit_df=...)` fits on the train slice and scores the test set.

extract → transform → **detect** → load

```python
# every detector, one contract
class Detector(ABC):
    name: str
    def fit(self, df) -> Self: ...
    def predict(self, df) -> Series:   # bool per row
        ...
    def score(self, df) -> Series:     # graded score
        return self.predict(df).astype(float)

# classical ML: adapt PyOD, do not reimplement
class PyODDetector(Detector):
    def fit(self, df):
        x = impute(df[self.numeric_cols])
        self.model = self.estimator.fit(scale(x))
        return self

# registry: one line, picked up everywhere
"isolation_forest": lambda: IsolationForestDetector(),
```

05 TWO FAMILIES NOW, A THIRD PLANNED

# Seven detectors, one contract

Classical ML detectors are wrapped from PyOD, not reimplemented. The flag threshold is read from the train-set prevalence. The time-aware `drift` detector is opt-in and runs on the synthetic dataset only. Deep learning (Autoencoder, LSTM-Autoencoder) is M2.

| Detector | Family | Principle | Main limit |
|---|---|---|---|
| Z-score | statistical rule | flag `\|z\| > 3` on any column | mean and std inflated by outliers |
| IQR | statistical rule | Tukey fences, `k = 1.5` | any-column rule fires often in high dim |
| Great Expectations | declarative rule | not-null + range at 0.1 / 99.9 pct | only checks what the suite declares |
| Isolation Forest | classical ML | isolation depth under random splits | axis-aligned splits, weak on nulls |
| Local Outlier Factor | classical ML | local density vs 20 neighbours | scoring cost grows with the scored set |
| One-Class SVM | classical ML | RBF boundary around the bulk | quadratic training |
| drift | time-aware | recent-window mean shift, per-row lean | per-row, so diluted drift is invisible |

06 FOUR DATASETS, ONE PROTOCOL

# Three real settings bracket the method

Stratified 70 / 15 / 15 split, fit on train, every number reported on the held-out test set. Five seeds, mean ± std. The ML contamination is read from the train-set prevalence, not fixed at 10%.

**0.01 → 0.20**

Isolation Forest fraud precision, from the contamination change alone

| Dataset | Rows | Feat. | Anom. | Role |
|---|---|---|---|---|
| Credit Card Fraud | 283,726 | 30 | 0.17% | rare, but a mixed-defect label |
| Cardio | 1,831 | 21 | 9.6% | rare and outlying, the fitting regime |
| Water Quality | 3,276 | 9 | 61% | target class common, wrong regime |
| Synthetic per-type | 12,360 | 9 | 17.5% | controlled, 6 labelled types |

LOF and One-Class SVM run everywhere except Credit Card Fraud, where neither scales to 284k rows in the pipeline.

07 HOW EACH TYPE IS INJECTED

# Magnitudes are a swept parameter, not a fixed choice

**severity = 1.0** is the reference dataset. Missing values, duplicates and invalid formats have no magnitude to scale. The severity sweep (slide 15) turns "we picked the magnitudes" into a curve and shows the per-type ranking is stable. Five types replace clean rows; duplicates are appended.

| | |
|---|---|
| MISSING VALUE | 1–2 of {income, balance, avg amount, tenure, n txn, email, country} → null |
| DUPLICATE | exact copy of a sampled clean row, id included |
| UNIVARIATE | one numeric column × U(15, 45) × severity |
| MULTIVARIATE | marginals at the (1 − 0.03 / severity) percentile of the clean data |
| TEMPORAL DRIFT | signup in the last 80 days; income × (1 + 0.55 · severity), similar for balance and amount |
| INVALID FORMAT | email → malformed; age → impossible; country → out of domain |

At `severity = 1` these are the reference magnitudes. Sweeping the multiplier (slide 15) shows every distributional detector's recall on univariate, multivariate and drift rises smoothly, and the ranking does not change.

*Aggregate metrics say the detector families are interchangeable.* Per anomaly type, they are not.

F1 spans 0.31 to 0.51 on the synthetic set · the per-type competences barely overlap

# The operating point decides the outcome

The statistical rules recover most frauds at precision below 0.04. Isolation Forest changes once its contamination matches the prevalence: it flags 0.17% of rows and reaches the first usable precision, at recall 0.21 and with real seed variance (F1 = 0.20 ± 0.07). Great Expectations leads PR-AUC.

**0.20**

Isolation Forest precision (was 0.01 at contamination 0.10)

**0.182**

Great Expectations PR-AUC, the best on this dataset

| Detector | Prec | Rec | F1 | ROC | PR | Flag% | Time |
|---|---|---|---|---|---|---|---|
| Z-score | 0.011 | 0.870 | 0.021 | 0.926 | 0.078 | 13.4 | 0.03s |
| IQR | 0.003 | **0.966** | 0.007 | 0.911 | 0.018 | 48.6 | 0.12s |
| Great Expectations | 0.037 | 0.676 | 0.070 | 0.830 | **0.182** | 3.1 | 1.37s |
| Isolation Forest | **0.202** | 0.206 | **0.203** | **0.934** | 0.120 | 0.2 | 1.78s |

Mean over 5 seeds, 15% held-out test split. LOF and One-Class SVM do not scale to 284k rows here.

# When anomalies are rare and outlying, the learned detectors work

Isolation Forest reaches F1 0.62 and PR-AUC 0.70. Z-score also does well: the 21 features are standardized and anomalies stand out on several at once. This is the counterpoint to a 0.17% mixed label and a 61% majority.

**0.62**
Isolation Forest F1 on Cardio

**0.70**
Isolation Forest PR-AUC on Cardio

| Detector | Prec | Rec | F1 | ROC | PR | Flag% |
|---|---|---|---|---|---|---|
| Z-score | 0.49 | **0.82** | 0.61 | 0.93 | 0.61 | 16.6 |
| IQR | 0.21 | 0.90 | 0.34 | 0.78 | 0.30 | 43.4 |
| Great Expectations | 0.51 | 0.21 | 0.29 | 0.59 | 0.19 | 4.6 |
| Isolation Forest | **0.64** | 0.62 | **0.62** | **0.95** | **0.70** | 9.9 |
| LOF | 0.25 | 0.27 | 0.26 | 0.62 | 0.24 | 10.7 |
| One-Class SVM | 0.56 | 0.61 | 0.58 | 0.94 | 0.63 | 10.7 |

Cardiotocography, 1,831 rows, 9.6% anomalies. Mean over 5 seeds, test split. Wide seed spread on this small set (ML F1 std up to 0.17).

# Aggregate metrics make the families look interchangeable

Isolation Forest has the best F1; no family is clearly ahead. Z-score keeps a perfect precision by firing only on the most extreme rows. One-Class SVM matches Isolation Forest at about 16× the latency.

**0.31–0.51**

F1 range across all six detectors

**0.33–0.56**

PR-AUC range: no clear winner

| Detector | Prec | Rec | F1 | ROC | PR | Time |
|---|---|---|---|---|---|---|
| RULE-BASED | | | | | | |
| Z-score | **1.000** | 0.182 | 0.308 | **0.748** | **0.560** | 0.002s |
| IQR | 0.603 | 0.399 | 0.481 | 0.678 | 0.460 | 0.004s |
| Great Expectations | 0.920 | 0.205 | 0.335 | 0.601 | 0.332 | 0.29s |
| CLASSICAL ML | | | | | | |
| Isolation Forest | 0.515 | **0.503** | **0.509** | 0.745 | 0.501 | 0.16s |
| Local Outlier Factor | 0.380 | 0.390 | 0.385 | 0.646 | 0.437 | 0.14s |
| One-Class SVM | 0.488 | 0.478 | 0.483 | 0.729 | 0.531 | 2.61s |

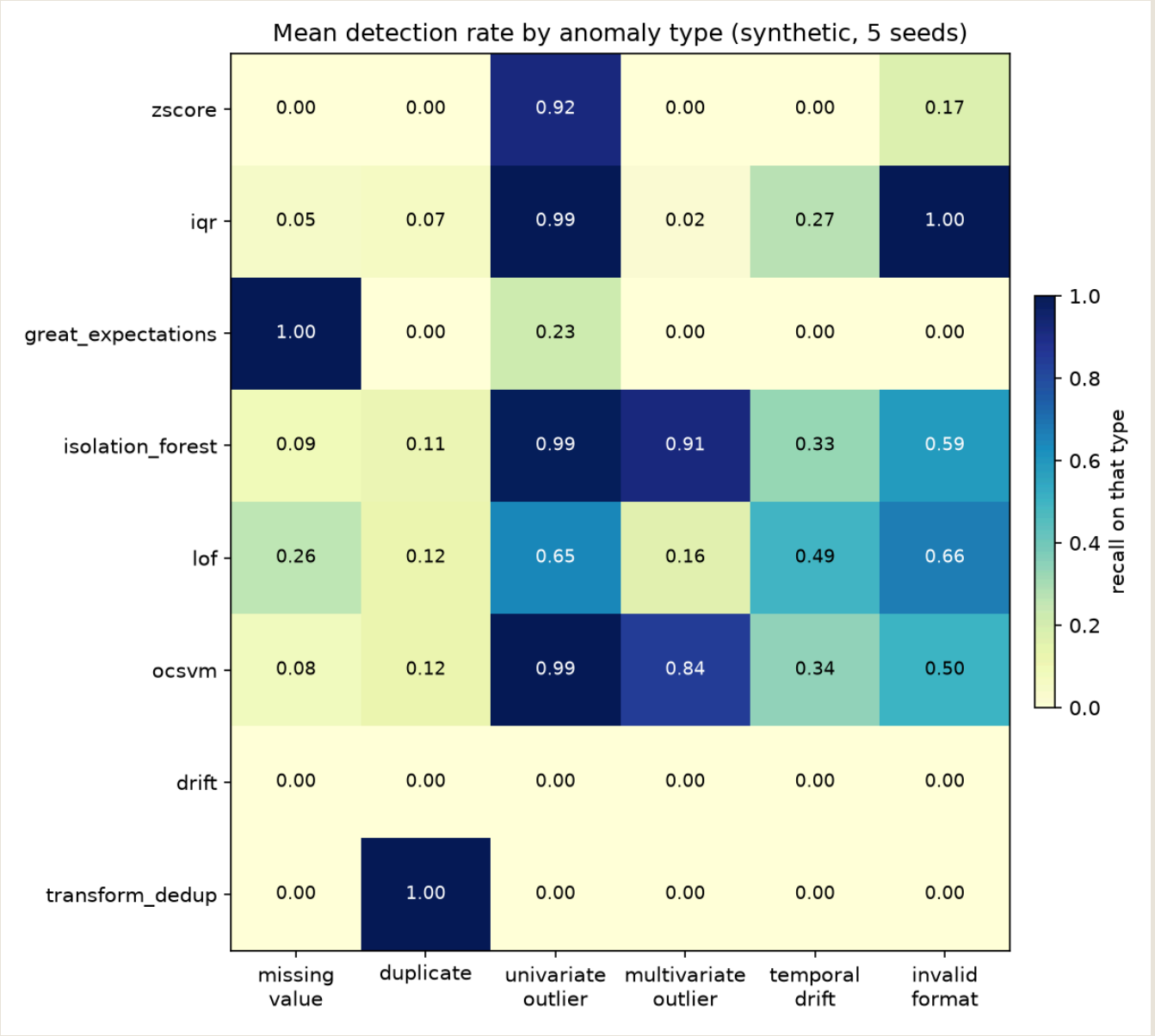
Mean over 5 seeds, test split. 12,360 rows, 17.5% anomalous.

# Each method has a narrow competence

Z-score owns univariate outliers. Great Expectations owns missing values (recall 1.00, precision 0.90). Deduplication owns exact duplicates. Isolation Forest and One-Class SVM are the only methods that catch multivariate outliers. Broad coverage needs several families, not the single best one.

## 0.91 vs ≤0.02

Isolation Forest recall on multivariate outliers, against every rule




paper/figures/heatmap_synthetic.png

transform_dedup and drift shown for reference

13 RECALL AND PRECISION, PER TYPE

# Detection rate alone can be gamed

Isolation Forest catches 91% of multivariate outliers at only 0.24 precision: a wide net that also pulls in clean rows. Z-score fires only when it is certain, so its precision is 1.00 on the two columns it covers.

**0.91 / 0.24**

recall / precision on multivariate outliers, Isolation Forest

| Method | missing | | univ. out. | | multiv. out. | | drift | |
|---|---|---|---|---|---|---|---|---|
| | rec | prec | rec | prec | rec | prec | rec | prec |
| Z-score | 0.00 | – | **0.92** | **1.00** | 0.00 | – | 0.00 | – |
| IQR | 0.05 | 0.03 | **0.99** | 0.39 | 0.02 | 0.01 | 0.27 | 0.15 |
| Great Expectations | **1.00** | **0.90** | 0.23 | 0.70 | 0.00 | – | 0.00 | – |
| Isolation Forest | 0.09 | 0.03 | **0.99** | 0.26 | **0.91** | 0.24 | 0.33 | 0.10 |
| LOF | 0.26 | 0.06 | 0.65 | 0.15 | 0.16 | 0.04 | **0.49** | 0.11 |
| One-Class SVM | 0.08 | 0.03 | **0.99** | 0.25 | 0.84 | 0.22 | 0.34 | 0.10 |
| transform dedup | 0.00 | – | 0.00 | – | 0.00 | – | 0.00 | – |

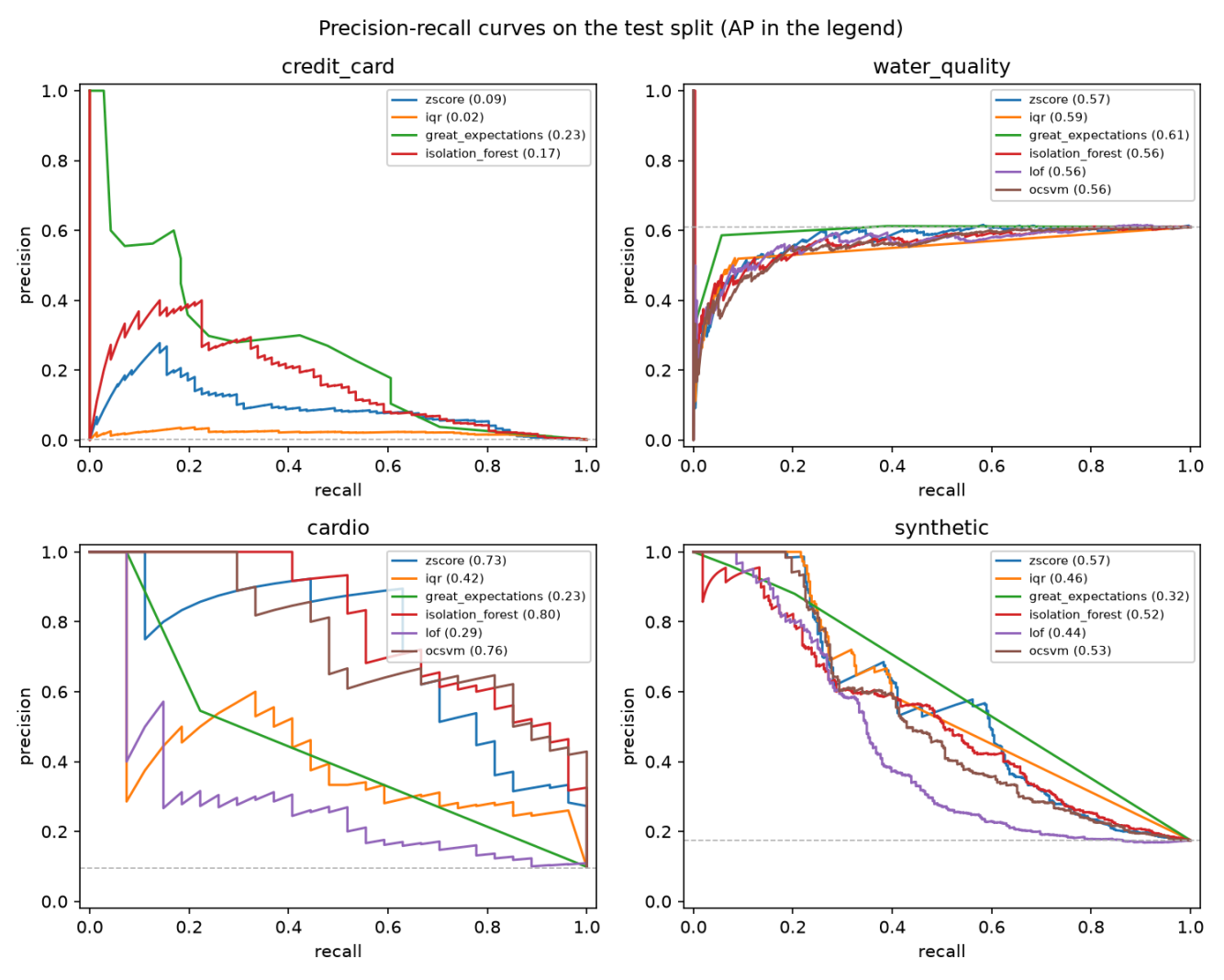
Recall and precision per anomaly type, clean rows as the only negatives. 5 seeds, mean. Duplicate and invalid-format columns omitted for space.

# One threshold hides the whole trade-off

On Cardio the Isolation Forest and One-Class SVM curves stay high (AP 0.80, 0.76). On Water Quality every curve hugs the 0.61 base-rate line. On Credit Card Fraud two methods dominate but neither clears precision 0.4 except at negligible recall.

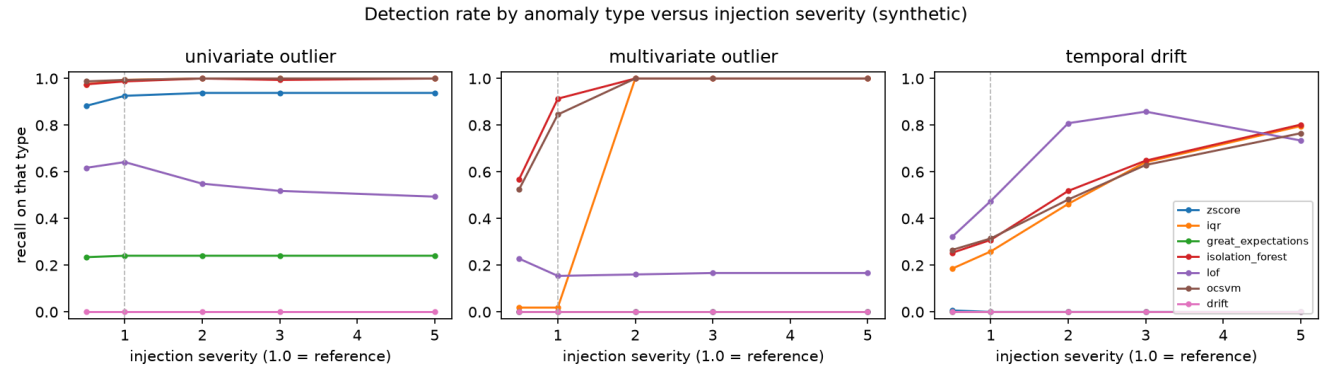
## 0.37 → 0.008

Isolation Forest fraud precision as recall goes 0.23 → 0.92



# The per-type ranking is stable as the magnitudes scale

Univariate outliers are saturated already at severity 0.5. Multivariate outliers show a threshold. Temporal drift climbs roughly linearly for the distributional detectors: the reference drift is simply below every method's floor, not a property of the methods. The same families lead the same columns at every severity.




**≤0.3 → 0.85**

temporal-drift recall from severity 1 to 3

16 THE PRACTICAL OUTPUT

# Anomaly type to method

Two of six types are not a per-row detector problem at all: duplicates belong to the transform stage, drift to a windowed monitor. That itself is a finding a single aggregate ranking would never surface.

| Anomaly type | Use | Recall | Why |
|---|---|---|---|
| missing value | Great Expectations | 1.00 | not-null expectation, exact |
| exact duplicate | transform dedup stage | 1.00 | row equality, not a detector's job |
| univariate outlier | any statistical rule | 0.92–0.99 | single-column band is enough |
| multivariate outlier | Isolation Forest | 0.91 | only the joint models see it (precision 0.24) |
| temporal drift | windowed monitor | ≤0.49 | sub-threshold per row; needs a stage |
| invalid format | IQR, partially | 1.00 / none | numeric part caught, string format missed |

# Unsupervised detection works in one of three regimes

The question is less "rules or ML" than "does this dataset have rare, outlying anomalies at all". Latency and scaling then constrain which detector can sit in a real flow: Great Expectations is the slowest linear method, and the One-Class SVM and LOF do not survive the full fraud dataset.

## Fraud

### too extreme

0.17% positives, and the label mixes every defect. No method reaches a usable precision at reasonable recall.

## Cardio

### the fit

9.6%, rare and distinct. Isolation Forest F1 0.62, PR-AUC 0.70. The regime the detectors are built for.

## Water

### wrong regime

61% majority target. Every ROC-AUC 0.45 to 0.51: the detectors are at chance.

The synthetic dataset sits in the middle at 17.5% and is where the per-type structure is measured. Pipeline cost narrows the shortlist further: Great Expectations is the slowest linear method, and the One-Class SVM and LOF do not survive the full fraud dataset.

# Findings from the build

Each of these became a paragraph in the paper. The recurring lesson: the setup around a detector, the operating point, the regime, the pipeline cost, matters as much as the algorithm.

### THE OPERATING POINT IS A LEVER

Contamination = prevalence instead of a fixed 10% moves Isolation Forest fraud precision from 0.01 to 0.20 and multivariate recall from 0.41 to 0.91.

### ONE REGIME OUT OF THREE

Cardio: F1 0.62. Fraud (mixed label): nothing usable. Water Quality (61% majority): at chance.

### GX 1.X IS A FULL API REWRITE

Per-row failures need the COMPLETE result format plus an explicit index column. Figured out by probing the installed library.

### ONE-CLASS SVM DOES NOT SCALE

Quadratic training, and LOF scoring is superlinear. On the full fraud dataset only rules and Isolation Forest run.

### ML IS WEAK ON MISSING VALUES

Imputing removes the signal a not-null rule catches for free: GX recall 1.00 against 0.08 to 0.26 for the learned models.

### DRIFT IS NOT A PER-ROW PROBLEM

Once diluted by clean recent rows the window mean has moved by about a tenth of a standard deviation.

19 IMPLEMENTED AND TESTED

# What the repository holds today

Every module has behavioural tests: detector outputs, split balance, contamination flow, synthetic generator, benchmark aggregation. conda env, Makefile, versioned paper PDF, CI on push.

**82**
tests, ruff clean, CI green

**8**
commits since the last review

| Module | Tests | Covers |
|---|---|---|
| quality/rules, gx | 13 | flags, constant column, score, null + range |
| quality/ml | 16 | 3 detectors: planted outliers, ranking, imputation |
| quality/drift | 5 | window shift, no-time no-op, index |
| split, registry, datasets | 16 | ratios, class balance, contamination flow |
| synthetic | 12 | reproducible, all types, severity monotone |
| run_benchmark, run_synthetic_benchmark | 8 | mean ± std over seeds, per-type precision, figure |
| pipeline | 4 | label excluded, fit-sample, timings |

20 WHERE IT STANDS

# Rigour gaps closed, substance left for M2

Every rigour gap from the last review is closed: held-out split, seeds, contamination, per-type precision, an imbalanced real dataset, CI. Deep learning and the write-up are the work left for M2. Defense July 2027.

| | |
|---|---|
| held-out split, 5 seeds, mean ± std | DONE |
| contamination from prevalence, full sweep | DONE |
| per-type precision, stable under severity | DONE |
| imbalanced real dataset (Cardio) | DONE |
| CI: GitHub Actions, ruff + pytest | DONE |
| deep learning: Autoencoder, LSTM-AE | M2 |
| windowed drift monitor | OPEN |
| Related Work, bibliography, final Discussion | M2 |

# Rules and machine learning are complementary by anomaly type.

The per-type heatmap is what makes that choice actionable. Across three real datasets, unsupervised detection pays off only on Cardio, where the anomalies are genuinely rare outliers. The pipeline, the four datasets, the held-out protocol, the curves and the CI are in place and tested. Deep learning and the write-up are the work for M2.